The Nation as Identity of Last Resort

Duk Gyoo Kim

Yonsei University

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Very Preliminary

The question

- **The classic puzzle:** Why do many low-income citizens identify with the nation and demand *less* redistribution than their material interest dictates?
- **Standard account (Shayo, 2009):**
  - Discrete classes (Poor, Rich); agents trade off *exogenous* national status against a *quadratic distance loss* from the group's average member.
  - Only the poor switch to national identity to borrow standing, depressing redistribution.
- **What happens when we consider endogenous status?**
  1. If the status of national identity depends on *who visibly claims it*, low-income identifiers dilute the very standing they seek to borrow. How is this equilibrium sustained?
  2. If the poor abandon class for the nation, **who actually sustains redistribution**?

Preview of main results

1. **The Nation as "Identity of Last Resort":** High-status clubs screen out low-income entrants to prevent status dilution. The nation cannot (**income-non-excludability**), allowing the poor to claim it despite imposing a negative composition externality.
2. **Two-tail sorting with a middle hold-out:** Unlike quadratic distance (where the middle is closest to the mean), balancing borrowed standing against local distinctiveness yields national identifiers at **both income tails**, while the middle class retains class identity.
3. **The middle class sustains redistribution for the poor:** In majority voting, national salience lowers preferred taxes. Because low-income identifiers demand less redistribution, **it is the class-salient middle hold-outs who sustain the tax rate**.
4. **Revealed by counterfactual screening:** Raising an income eligibility floor raises national status and draws the middle class into national identity—which **lowers the majority tax and makes low-income citizens materially worse off**.

Motivating observation: Visible low-income identifiers



Taegeukgi rally at Seoul Plaza, Mar. 1, 2017.

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- South Korea's *Taegeukgi* (national flag) ralliers: conservative alignment tied to an encompassing national identity (security, Korea–US alliance) rather than economic class.
- Evidence shows a substantial portion are low-income elderly voters (Lee and You 2019; Lee 2021; Kim, Kang, and Ham 2021).
- **Observer bottom-weighting ($\omega(\theta)$):** Even when some wealthy voters participate, outsiders' perception of the visible identifier community disproportionately weights its low-income members.
- **The theoretical puzzle:** How is an identity equilibrium sustained when low-income members dilute the group's standing, yet the group lacks an institutional gatekeeper to exclude them?

Departure from Shayo (2009), If Time Permits

Dimension	Shayo (2009)	This Paper
Type space	Discrete classes (Poor, Rich)	Continuum $\theta \in [\underline{\theta}, \overline{\theta}]$
Who defines national status?	Entire population Θ (exogenous status)	**Visible identifiers** $\mathsf{N} \subseteq \Theta$: bottom-weighted $s(\mathsf{N})$ + eminence a
Composition externality	None (non-identifiers still anchor the national prototype)	Low-income entrants ($\theta < s$) dilute $s(\mathsf{N})$; no gatekeeper (**non-excludability**)
Identity trade-off	Cognitive distance: quadratic loss from group mean $-\gamma(\theta - \mu)^2$	Optimal Distinctiveness (Brewer 1991): borrowed status $\beta(a + s)$ vs. $k(\theta) + \delta(\theta)$
Who identifies nationally?	Single low tail (or middle if closest to μ)	**Two tails** $[\underline{\theta}, \underline{\tau}] \cup [\overline{\tau}, \overline{\theta}]$; middle holds out to preserve $\delta(\theta)$

Environment: Stage 1 (Identity formation)

- Citizen's income $\theta \in \Theta = [\underline{\theta}, \overline{\theta}] \subset \mathbb{R}_{++}$, distribution F, continuous density $f > 0$.
- Vertically nested memberships: each citizen chooses salient identity $g \in \{c, n\}$. Write $\mathsf{N} \subseteq \Theta$ for the set of national identifiers ($g = n$).
- Standing of national identity = $a + s(\mathsf{N})$:
  - $a > 0$: **inherited (pre-tax) national eminence** $a \equiv a(0)$ (international standing, economic weight; *in Stage 2, taxation t will be perceived to erode this standing*).
  - $s(\mathsf{N}) = \frac{\int_{\mathsf{N}} \omega(\theta)\theta \, dF(\theta)}{\int_{\mathsf{N}} \omega(\theta) \, dF(\theta)}$: endogenous **composition status**, with $\omega(\theta) > 0$ non-increasing (observers weight the bottom, e.g., $\omega_\xi(\theta) = e^{-\xi\theta}$).
- Identity payoffs (Optimal Distinctiveness, Brewer 1991):

$$U_n(\theta; \mathsf{N}) = \beta(\theta)\,[a + s(\mathsf{N})] - k(\theta), \qquad U_c(\theta) = \delta(\theta).$$

  - $\beta(\theta) > 0$ decreasing: the poor place the highest marginal value on borrowed standing.
  - $k(\theta) > 0$ decreasing: submersion cost; $\delta(\theta) > 0$: retained class distinctiveness.
- Net gain: $\Delta(\theta; a, s) \equiv \beta(\theta)(a + s) - k(\theta) - \delta(\theta)$, with $\partial\Delta/\partial s = \beta(\theta) > 0$.

The composition externality and the institution

- **Lemma 2 (Marginal composition externality):** Add a marginal mass ε of identifiers concentrated near type $z \notin \mathsf{N}$:

$$\left.\frac{ds(\mathsf{N}_\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = \frac{\omega(z)}{\int_{\mathsf{N}} \omega(\theta) dF(\theta)}(z - s).$$

  - Entry by $z < s$ strictly lowers status s, hurting incumbent θ by $\beta(\theta)\frac{\omega(z)}{W}(z - s) < 0$.
  - Hurt is largest for the poorest incumbents (largest β). No individual internalizes it.
- Clubs and professions prevent dilution via entry screening; national identity doesn't. Counterfactual benchmark: an **income-contingent floor** $\phi \in [\underline{\theta}, \overline{\theta}]$ admitting $\theta \geq \phi$.
- **Income-non-excludability** is the baseline institution $\phi = \underline{\theta}$: conditional on formal citizenship, no gatekeeper can bar a poor citizen from claiming national identity.
- ϕ is an institutional parameter for comparative statics, not a policy prescription.

Equilibrium existence and structure

- Given floor ϕ and candidate status s:

$$\mathsf{N}_\phi(s) = \{\theta \geq \phi : \Delta(\theta; a, s) \geq 0\}, \qquad \Phi_\phi(s) = s(\mathsf{N}_\phi(s)).$$

- A floor ϕ is **admissible** if $\inf_{s\in\Theta} \int_{\mathsf{N}_\phi(s)} dF(\theta) > 0$ (keeps denominator > 0).
- $\Phi_\phi(s)$ **may not be monotone:** Higher s expands $\mathsf{N}_\phi(s)$ at both margins, admitting new low types alongside high ones, which can pull the ω-weighted mean down.
- **Assumption 1 (Regularity):**
  - (i) Indifferent set $\{\theta : \Delta(\theta; a, s) = 0\}$ has F-measure zero for every $s \in \Theta$ (ensures $\Phi_\phi(s)$ is continuous via dominated convergence).
  - (ii) $\Delta(\underline{\theta}; a, \underline{\theta}) > 0$: poorest types identify even at the lowest possible status (ensures $\phi = \underline{\theta}$ is admissible).

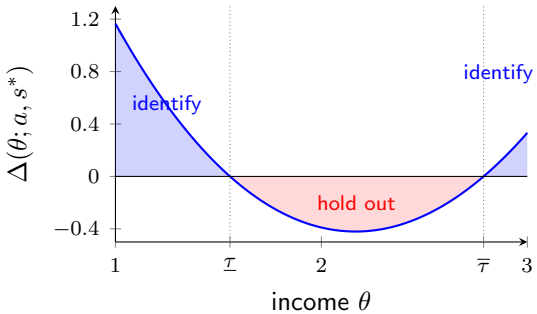
Proposition 1 (Existence): Under Assumption 1(i), an identity equilibrium $\Phi_\phi(s^*) = s^*$ exists for any admissible ϕ. The equilibrium set is compact, so $s^*_{\max}(\phi)$ is well-defined.

Sorting: Three equilibrium configurations

- **Assumption 2 (Two roots):** $\Delta(\cdot; a, s)$ has at most two zeros on Θ.
- **Lemma 1 (Primitive sufficiency):** If $\beta(\theta)(a+s) - k(\theta)$ is convex and $\delta(\theta)$ is concave, then $\Delta(\cdot; a, s)$ is convex $\Rightarrow$ Assumption 2 holds.

Proposition 2 ($\phi = \underline{\theta}$): At any equilibrium s^*, one of three cases holds (up to F-null sets):

1. **Pooling**: $\mathsf{N}^* = \Theta$.
2. **Single low tail**: $\mathsf{N}^* = [\underline{\theta}, \tau]$ (Shayo, 2009).
3. **Two tails**: $\mathsf{N}^* = [\underline{\theta}, \underline{\tau}] \cup [\overline{\tau}, \overline{\theta}]$, with middle hold-outs $(\underline{\tau}, \overline{\tau})$.



Two-tail equilibrium in the worked example $(a = 0.7, \xi = 2, s^* = 1.215)$.

- **Opposite motives in Case (3):** Poor identify to borrow standing ($\beta(a + s^*)$ high); rich identify because opportunity cost $k(\theta) + \delta(\theta)$ fades; middle has the most class distinctiveness $\delta(\theta)$ to lose.

Running parametric example: Role of national eminence a

$\Theta = [1, 3]$, $f(\theta) = \frac{3}{2} - \frac{\theta}{2}$ (right-skewed: $\theta_{\text{med}} \approx 1.586 < \mu \approx 1.667$), $\beta(\theta) = \frac{2}{\theta+1}$, $k(\theta) = \frac{0.5}{\theta+1}$, $\delta(\theta) = 1.5 - (\theta - 2)^2$, $\omega_\xi(\theta) = e^{-\xi\theta}$ ($\xi = 2$). Those satisfy Assumptions 1–2.

National eminence a	Equilibrium configuration	Sorting pattern
$a < 0.148$	Single low tail	Only the poorest identify ($\Delta(\overline{\theta}) < 0$)
$0.148 < a < 1.238$	**Two tails**	Poor and rich identify; middle holds out
$a > 1.238$	Pooling	Every type identifies ($\mathsf{N}^* = \Theta$)

- **Baseline ($a = 0.7$):** Unique equilibrium $s^* = 1.2154$, cutoffs $\underline{\tau} = 1.556$, $\overline{\tau} = 2.788$.
- Masses: low tail 0.479 (42.6% of whom have $\theta < s^*$), hold-outs 0.510, upper tail 0.011.
- Both θ_{med} and μ lie inside the middle hold-out region.
- Raising a ($0.7 \to 0.8$) raises s^* ($1.215 \to 1.234$) and expands both tails ($\underline{\tau} \uparrow$ and $\overline{\tau} \downarrow$), the lower margin by more ($+0.069$ vs. -0.049).

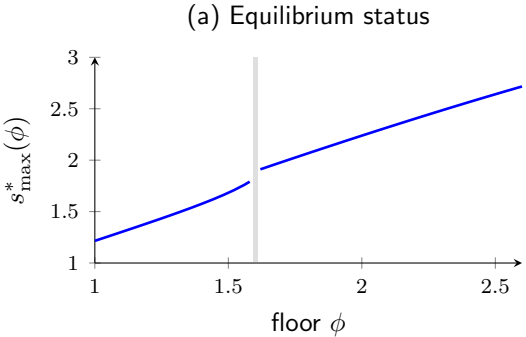
Counterfactual screening: Moving the floor ϕ

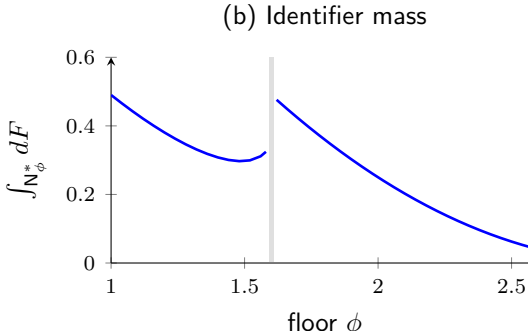
- Treat the income floor $\phi \in [\underline{\theta}, \overline{\theta}]$ as a counterfactual parameter ($\phi > \underline{\tau}$ mechanically removes the entire baseline low-income tail).
- **Lemma 4 (Monotone institutional comparative static):** For admissible floors $\phi' > \phi$:

$$\Phi_{\phi'}(s) \geq \Phi_{\phi}(s) \quad \forall s \in \Theta \quad \Longrightarrow \quad s^*_{\max}(\phi') \geq s^*_{\max}(\phi).$$

  Intuition: Truncating $\mathsf{N}_\phi(s)$ from below mechanically raises its ω-weighted mean; fixed-point comparison delivers monotone equilibrium status.
- **Induced entry of the middle:** Because $\partial\Delta/\partial s = \beta(\theta) > 0$, higher equilibrium status raises Δ for *every* remaining eligible type, drawing former middle hold-outs into N^*_ϕ.
- **Ambiguous total identifier mass:** Direct exclusion at the bottom vs. induced entry in the middle work in opposite directions for moderate ϕ (vanishes as $\phi \to \overline{\theta}$).

Moving the floor: Equilibrium status vs. identifier mass





- **Status $s^*_{\max}(\phi)$ rises monotonically** (Lemma 4).
- **Non-monotone identifier mass:** It falls from 0.490 to 0.297 ($\phi \approx 1.48$), jumps back to 0.490 across the fold band $\phi \in [1.593, 1.611]$ as the entire hold-out region nationalizes at once, then declines.

Stage 2: Redistribution and stewardship of national eminence

- **From Stage 1 (a) to Stage 2 ($a(t)$):**
  - In Stage 1, citizens sort into identities $g^*_\phi(\theta) \in \{c, n\}$ given **inherited national eminence** $a > 0$.
  - In Stage 2, citizens vote over a linear tax $t \in [0, 1]$ funding a lump-sum transfer $\mu = \mathbf{E}_F[\theta]$:

$$m(t; \theta) = (1 - t)\theta + t\mu - \tfrac{d}{2}t^2, \qquad d > 0.$$

- **Perceived fiscal erosion of eminence (p):** Citizens who adopt $g = n$ internalize a norm to preserve the standing a they borrow. Taxation t is perceived to erode a fraction pt of inherited eminence, leaving $a(t) = a(1 - pt)$, so national identifiers internalize the loss $a - a(t) = pat$:

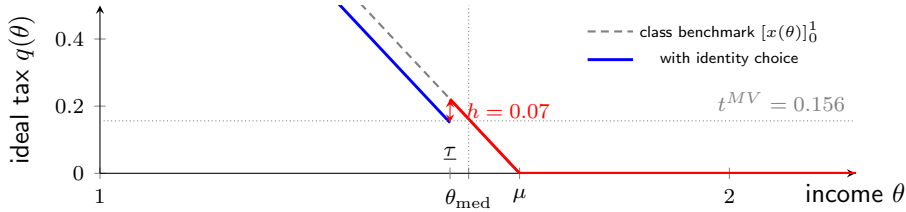
$$V(t; \theta, g) = m(t; \theta) - pat\mathbf{1}\{g = n\}.$$

  $p \in (0, 1]$ decouples non-economic prestige (a) from the perceived fiscal sensitivity of national standing.

- **Ideal tax rates:** With $x(\theta) \equiv \frac{\mu - \theta}{d}$ and $h(a) \equiv \frac{pa}{d}$,

$$t^*(\theta, c) = [x(\theta)]^1_0, \qquad t^*(\theta, n) = [x(\theta) - h(a)]^1_0.$$

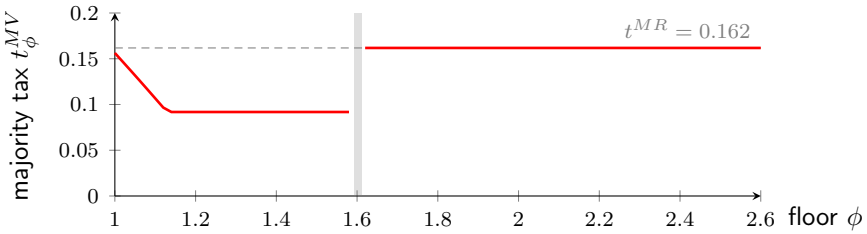
The majority tax and the role of the middle



Ideal tax schedule $q(\theta)$ at $a = 0.7$. Poorest types ($\theta \leq 1.132$) are clipped at $t = 1$ under both identities.

- Single-peaked $V(t; \theta, g) \Rightarrow$ majority tax t_ϕ^{MV} is the **median of the ideal-tax distribution**.
- At $\underline{\tau}$, $q_\phi(\underline{\theta}) = t^*(\underline{\theta}, g_\phi^*(\underline{\theta}))$ jumps *up* by $h(a)$ as identity switches from n to c.
- **Sustaining redistribution:** Class-salient hold-outs around θ_{med} hold $t^{MV} = 0.156$ above $x(\theta_{\text{med}}) - h = 0.092$. Poor national identifiers materially depend on them (Corollary 3).
- **How raising the floor ϕ shifts ideal taxes:**
  - **Directly excluded poor** revert to class $\Rightarrow$ ideal taxes *rise* (unless already clipped at $t = 1$).
  - **Newly nationalized middle hold-outs** switch to nation $\Rightarrow$ ideal taxes *fall*.
  - If the excluded poor are clipped at $t = 1$, only the second channel operates and t_ϕ^{MV} weakly falls.

Screening and the majority tax: Worked example



$a = 0.7$, $h(a) = 0.07$, Meltzer–Richard benchmark $t^{MR} = 0.162$.

- **Baseline ($\phi = 1$):** $t^{MV} = 0.156$; class-salient hold-outs around θ_{med} sustain redistribution.
- **Mild screening ($\phi \uparrow 1.14$):** s^* rises, $\underline{\tau}$ crosses θ_{med} (at $\phi = 1.061$), nationalizing pivotal hold-outs. Since excluded types in $[1, 1.132]$ are clipped at $t = 1$, t^{MV} falls.
- **Stringent screening ($\phi \geq 1.593$):** θ_{med} herself is excluded, and t^{MV} jumps back to t^{MR}.
- **Material harm:** (Corollary 3) The drop $0.156 \to 0.092$ makes every type $\theta \leq 1.589$—including the entire low tail of national identifiers—materially worse off.

National eminence decides whether the middle is decisive

- **Moderate eminence ($a = 0.7$):** Hold-out region $(1.556, 2.788)$ contains both $\theta_{\text{med}} = 1.586$ and $\mu = 1.667$.
  - Hold-outs below μ demand positive redistribution ($t^*(\theta, c) > 0$), so the class-salient middle **sustains the majority tax**, and both screening channels operate.
- **High eminence ($a = 1.0$):** Lower cutoff rises to $\underline{\tau} = 1.783 > \mu$.
  - Every remaining hold-out has income above the mean ($\theta > \mu$) and prefers $t = 0$ under *either* identity.
  - Switching a hold-out's identity no longer changes her tax vote; the ideal-tax schedule has no jump at $\underline{\tau}$, and the majority tax $x(\theta_{\text{med}}) - h(1) = 0.062$ is set within the low-income national tail.
- **Takeaway:** Same primitives, different national eminence, different redistributive politics. Whether the middle class sustains redistribution depends on a and on institutional excludability.

Testable implications and experimental design

H1 Two-tail identification: In the two-root parameter region, national identification is U-shaped across income (higher in low and high tails than in the middle).

H2 Asymmetric eminence response: An exogenous increase in a expands the low-income identifier interval more than the high-income interval when $\frac{\beta(\underline{\tau})}{|\Delta_\theta(\underline{\tau})|} > \frac{\beta(\overline{\tau})}{\Delta_\theta(\overline{\tau})}$ and $1 + \frac{ds^*}{da} > 0$. (holds in example: $0.560 > 0.393, ds^*/da \approx 0.188$).

H3 Screening and redistribution: An income floor suppresses low-income identification, moves total identification non-monotonically, and shifts preferred redistribution down by $h(a)$ for interior ideal points.

- **Companion experiment (not yet):** Independently manipulates induced income, eminence a, composition feedback, floor ϕ, and voting to separately identify $\beta(\theta)$ vs. $k(\theta) + \delta(\theta)$ at the two tails.

Related literature

- **Social identity and redistribution:** Akerlof–Kranton (2000); Shayo (2009, 2020); Klor–Shayo (2010); Sambanis–Shayo (2013); Dhami et al. (2021); Bonomi, Gennaioli, Tabellini (2021). *Contribution:* Endogenizes national status to visible identifier composition under income-non-excludability.
- **Why the poor do not expropriate the rich:** Meltzer–Richard (1981); Roemer (1998); Bénabou–Ok (2001); Cruces et al. (2013); Alesina, Stantcheva, Teso (2018). *Contribution:* Internal identity-salience mechanism generating a middle-sustained redistributive equilibrium.
- **Group formation under status concern and clubs:** Buchanan (1965); Moldovanu, Sela, Shi (2007); Bernard, Hett, Mechtel (2016); Staab (2024). *Contribution:* Vertically nested membership where the exclusionary screening instrument is institutionally absent.
- **Ends against the middle:** Epple–Romano (1996a, 1996b); Kim (2019). *Contribution:* Non-monotone ideal-tax schedule driven by composition-dependent identity sorting while preserving single-peakedness at the voting stage.

Conclusion

- National identity serves as the **identity of last resort** because low-income citizens can dilute its composition standing without being screened out on income.
- Treating the eligibility floor ϕ as a counterfactual parameter shows that screening raises equilibrium status and draws in middle hold-outs, moving total national identification and majority redistribution non-monotonically.
- Class-salient middle hold-outs politically sustain the redistribution received by low-income national identifiers, provided national eminence a is not so high that it nationalizes the middle.